# Does Model Complexity Actually Pay Off in Credit Risk?

## 1. The Cost Matrix: Recall vs. Accuracy

In the context of credit risk prediction, accuracy can be a somewhat inaccurate metric of success due to its disregard of the different costs associated with the two types of classification errors. In the vast majority of data sets regarding lending, the number of people who pay off their loans is greater than the number of people who have defaults. As a result, a classification algorithm is likely to have a high level of accuracy when making the right prediction about most people repaying their loans, but failing to recognize many defaulters.

In the context of banking, Recall for the Default class becomes very important as it is essentially the fraction of all defaulters who are detected by the classification algorithm. In case a bank does not detect a person who turns out to be a defaulter, the bank loses the loan amount in question. On the other hand, if the bank misclassifies a trustworthy client, it risks missing an opportunity to make some money.

## 2. Interpretation and Regulatory Barrier

Logistic Regression is considered interpretable as it estimates the effect of each borrower characteristic on the probability of default. Thus, its coefficients allow one to see whether there is a relation and how strong it is, allowing the bank to explain the reason why a particular applicant is risky. Such approach enables banks to create explanations of their decision-making process.

In its turn, the XGBoost algorithm uses a number of decision trees that find complicated and non-linear relations between borrower characteristics. Despite the possible increase in predictive performance, the resulting model becomes difficult to interpret as one cannot explain its decisions through the analysis of a single coefficient or rule.

To overcome this problem, banks have to make use of various explainability methods like SHapley Additive exPlanations (SHAP), which calculates the contribution of each feature to the output. Nevertheless, using XGBoost with SHAP requires additional engineering and operational efforts on the part of the bank. In particular, banks should maintain pipelines of explanations, validate

these explanations, monitor the model's behavior, and ensure the quality of the explanations.

On the other hand, even with SHAP, there is no automatic compliance with the regulatory standards. The banks have to make sure that the decisions that they are making are compliant with the fair lending, consumer protection, and adverse action laws. This can create additional complexities in the development process.

3. The Verdict

In the role of Chief Risk Officer in a retail lending bank, I would opt for a two-step hybrid solution, combining Logistic Regression with a sophisticated ensemble model wherever it makes sense.

On the one hand, I would implement Logistic Regression as a simple-to-understand baseline and validate its default detection accuracy, calibration, financial performance, and other characteristics. On the other hand, I would test whether XGBoost provides an actual boost in terms of default detection quality and expected credit loss minimization. If so, I would implement XGBoost model in parallel to SHAP-based explanations.

The proposed hybrid system would rely on the ensemble model for risk ranking but keep Logistic Regression as a transparent baseline or fall back whenever possible. Nevertheless, I wouldn't implement Logistic Regression for decision-making and XGBoost exclusively for risk assessment without thoroughly evaluating how those two steps interact. The design of the production process should guarantee that the model used for the lending decision is also explainable.

This method takes into account three key factors:

Defaulting identification: Finding out the borrowers who are likely to become defaulters.

Regulatory requirements: Being able to explain the decision and maintain proper documentation.

Operational convenience: Expenditure on infrastructure and monitoring should be justified only if there are any benefits.

In conclusion, it has to be noted that sometimes the most complicated algorithm does not have to be the optimal algorithm. It should be the one which brings significant benefit in terms of increased profitability adjusted for risk.